Spring, block and board

Y20 (2020) — English translation

Part A. Block on a smooth surface

A block of mass m rests on a smooth horizontal surface, connected to the wall by a weightless horizontal spring with a stiffness coefficient k, the deformation of which at the initial moment is zero. At the moment of time $t = 0$ the block begins to be acted upon by a constant force F directed from the wall.

A1	Find the maximum deformation of the spring ΔL_1 and the time t_1 after which it is reached.	**0.80**

A2	Find the maximum speed of the block v_{max1} and the time t_2 after which it is achieved.	**0.80**

A3	Find the maximum acceleration of the block a_{max1} and the minimum time t_3 after which it is achieved.	**0.40**

Part B. Block on board

A board of mass M rests on a smooth horizontal surface, a block of mass m rests on the board, the coefficient of friction between the block and the board is μ. The block is connected to the wall by a horizontal spring with a stiffness coefficient k, the deformation of which at the initial moment is zero. At time $t = 0$, a constant force F, directed from the wall, begins to act on $доску$. The acceleration due to gravity is g. Consider the board to be infinitely long.

B1	Find the maximum deformation of the spring ΔL_2 and the time t_4 after which it is reached.	**3.00**

B2	Find the maximum speed of the block v_{max2} and the time t_5 after which it is achieved.	**2.50**

B3	Find the maximum acceleration of the block a_{max2} and the minimum time t_6 after which it is achieved.	**1.50**

Part C. Cylinder on board

A board rests on a smooth horizontal surface, a solid homogeneous cylinder rests on the board, the coefficient of friction between the cylinder and the board is μ. The cylinder is connected to the wall by two horizontal springs with a stiffness coefficient k, the deformation of which at the initial moment is zero. The attachment points for the springs to the cylinder are located on its axis. At time $t = 0$ the board begins to move with constant acceleration a directed from the wall. The acceleration due to gravity is g. Consider the board to be infinitely long.

C1	Find the maximum deformation of the spring ΔL_3 and the time t_7 after which it is reached.	**1.50**

C2	Find the maximum speed of the center of the cylinder v_{maxc} and the time t_8 after which it is reached.	**1.00**

C3	Find the maximum acceleration of the center of the cylinder a_{maxc} and the minimum time t_9 after which it is achieved.	**0.50**

Source: https://pho.rs/p/45